

Nitrogen (N₂)

A colorless, odorless, nonflammable gas — or a colorless, odorless, nonflammable cryogenic liquid.

GRADES & SPECIFICATIONS

Grade	% Purity	O ₂	H ₂ O	THC	CO	CO ₂
Research Ultra 6.0	99.9999	10 ppb	20 ppb	100 ppb	—	—
Research Plus 6.0	99.9999	0.2	0.2	0.1	0.3	0.1
Research 5.7	99.9997	0.5	0.5	0.2	0.5	0.5
Semiconductor 5.3	99.9993	1	1	0.5	1 ⁽¹⁾	1 ⁽¹⁾
Ultra-Pure Carrier (UPC)	99.9993	1	1	0.5	1 ⁽¹⁾	1 ⁽¹⁾
Ultra High Purity (UHP) 5.0	99.999	1	1	0.5	1 ⁽¹⁾	1 ⁽¹⁾
High Purity (HP)	99.998	5	—	—	—	—

Concentrations given are ppm by volume unless otherwise specified.

Purities do not include Argon impurities @ <100 ppm. **(1)** Combined CO and CO₂ shall not exceed 1 ppm.

Individual or Batch Certificate of Analysis available upon request.

ORDERING INFORMATION

Product	Cyl. Size	Contents (ft ³)	Valve (CGA)	Product Number	Pressure @70°F (psig)	Delivery Range (psig)
Research Ultra 6.0	300	304	580	NI-RU-300	2,640	2-stage · 2–75
Research Plus 6.0	300	304	580	NI-RP-300	2,640	2-stage · 0–150
	200	235	580	NI-RP-200	2,200	
	80	93	580	NI-RP-80	2,200	
	35	36	580	NI-RP-35	2,200	
Research 5.7	300	304	580	NI-RES-300	2,640	Single / 2-stage
	200	235	580	NI-RES-200	2,200	
	80	93	580	NI-RES-80	2,200	
	35	36	580	NI-RES-35	2,200	
Semiconductor 5.3	300	304	580	NI-SEMI-300	2,640	2-stage
	200	235	580	NI-SEMI-200	2,220	
Ultra-Pure Carrier (UPC)	300	304	580	NI-UPC-300	2,640	Single-stage
	200	235	580	NI-UPC-200	2,200	
	80	93	580	NI-UPC-80	2,200	
	35	36	580	NI-UPC-35	2,200	

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Nitrogen (N₂)

Continued — Ordering Information & Technical Data

— ORDERING INFORMATION (CONT.)

Product	Cyl. Size	Contents (ft ³)	Valve (CGA)	Product Number	Pressure @70°F (psig)	Delivery Range (psig)
Ultra High Purity (UHP) 5.0	300	304	580	NI-UHP-300	2,640	Single / 2-stage · 0–100
	200	235	580	NI-UHP-200	2,200	
	80	93	580	NI-UHP-80	2,200	
	35	36	580	NI-UHP-35	2,200	
High Purity (HP)	6K	482	677	NI-HP-6K	6,000	200–3,500
	3K	337	680	NI-HP-3K	3,500	300–1,500

— TECHNICAL DATA & SHIPPING INFORMATION

Molecular Weight	28.0134 g/mol
Specific Volume	13.81 cf/lb @ 70°F & 1 ATM
Boiling Point	–195.8 °C
Flammability in Air	Nonflammable

U.S. DOT Name	Nitrogen, Compressed
ID Number	UN 1066
DOT Hazard Class	2.2 — Nonflammable Gas
CAS Registry Number	7727-37-9

WSM offers a wide range of gas supply modes, including high-pressure cylinders, liquid cylinders (dewars), MicroBulk and bulk. Your WSM specialty gas representative can help you select the option best suited to your operation, including automatic changeover systems for uninterrupted supply.

WSM Quality Policy. The purpose of the WSM Quality System is to continually improve our manufacturing and related processes to provide our customers with the highest product purity, consistency, and service.